Department: Computer Science and Engineering (Cyber Security)

Vision of the Department

"To ignite a passion for innovation in Computer Science and Cyber security, empowering students to become leaders who shape a secure and interconnected future"

Mission of the Department

- To provide an exceptional educational experience fostering critical thinking, creativity, and a deep understanding of cyber security principles.
- To equip graduates with in-demand skills and an ethical mindset to thrive in the dynamic cyber security landscape.
- To cultivate a collaborative learning environment connecting students with industry experts, cutting-edge research, and real-world challenges.
- To inspire a lifelong commitment to learning and dedication to building a safer, more secure digital world

Program Educational Objectives (PEOs)

- Excel as innovative technical leaders in computer science and cyber security, developing secure and robust solutions.
- Demonstrate a strong ethical mindset and contribute positively to a safer digital world.
- Show a continuous commitment to professional growth and actively engage with industry and research.
- To work efficiently in multi-disciplinary teams to establish
- Good soft skills to prove themselves at IT sector.

Program Outcomes (POs)

1. **Engineering Knowledge**: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
2. **Problem Analysis**: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3. **Design/Development of Solutions**: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
4. **Conduct Investigations of Complex Problems**: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
5. **Engineering Tool Usage**: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6. **The Engineer and The World**: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
7. **Ethics**: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
8. **Individual and Collaborative Team work**: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9. **Communication**: Communicate effectively and inclusively

within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

10. **Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and a leader in a team, and to manage projects and in multidisciplinary environments.

11. **Life-Long Learning**: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) Critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

1. Able to apply appropriate techniques for storage of huge amount of data and ensuring its integrity.
2. Choose appropriate method for data acquisition from real world and propose suitable solutions to solve problems.

SRI SIDDHARTHA ACADEMY OF HIGHER EDUCATION (SSAHE)

(Deemed to be University, Accredited A+ Grade by NAAC)

SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMAKURU-572105

(A Constituent College of SSAHE, Accredited by NBA)

Scheme of Teaching and Examination for NEP 160 Credits (2022 Scheme)

7TH SEMESTER **ACADEMIC YEAR: 2026 - 27**

Sl. No.	Course Code		Course Title	Teaching Dept.	Teaching Hrs.					Credits	Examination			
					L	T	P	S	Total		CIE Marks	SEE Marks	Total Marks	Exam Hrs
1	PC	22CY701	Cloud Computing and Security	CY	42	-	28	50	120	4	50	50	100	3
2	PC	22CY702	Digital Forensic and Incident Response	CY	42	-	28	50	120	4	50	50	100	3
3	PE	22CY7PE3X	Professional Elective-III	CY	42	-	-	48	90	3	50	50	100	3
4	PE	22CY7PE4X	Professional Elective-IV	CY	42	-	-	48	90	3	50	50	100	3
5	PE	22CY7PE5X	Professional Elective-V	CY	42	-	-	48	90	3	50	50	100	3
6	PE	22CY7PE6X	Professional Elective-VI	CY	42	-	-	48	90	3	50	50	100	3
7	PW	22CYPW707	Project Phase-I	CY	-	-	28	32	60	2	50	-	50	3
				Total	252		84	324	660	22	350	300	650	-

L: Lecture, T-Tutorial, P-Practical/Drawing, CIE: Continuous Internal Evaluation, SEE: Semester End Examination, S: Self Learning. XX: Branch Code

Credit Distribution: PC: Professional Core-08, PE: Professional Elective-12, PW: Project Work-02, Total=22 Credits.

Professional Elective III		Professional Elective IV		Professional Elective V		Professional Elective VI	
22CYPE731	Data Mining and Machine Learning	22CYPE741	Information Retrieval for Network Security	22CYPE751	Introduction to Deep Learning	22CYPE761	Cyber Forensics
22CYPE732	Cyber Threat Intelligence	22CYPE742	Malware Analysis and Reverse Engineering	22CYPE752	Digital and Mobile Forensic	22CYPE762	Social Network Security
22CYPE733	Web Application Security	22CYPE743	Distributed Systems	22CYPE753	Cognitive IoT	22CYPE763	Quantum Computing

Department: CSE (Cyber Security)			Semester:	VII
Subject: Cloud computing and Security				
Subject Code:	22CY701		L – T – P – S:	42 – 0 – 28 – 50 (Hrs)
			Credits:	4

	Course Objectives
1	Apply fundamental concepts of Cloud Computing
2	Analyze technological concepts of Cloud Computing.
3	Apply security standards, policies and compliance requirements to cloud applications.
4	Implement a resource scheduling algorithm for a given scenario using a suitable simulator.

Unit	Description	Hrs
I	**Introduction to Cloud Computing:** Introduction, Characteristics of cloud computing, Cloud Models: Service Models, Deployment Models, Cloud Services Examples: IaaS, PaaS, Saas, Cloud based services and applications, Cloud Concepts and Technologies.	09
II	**Virtualization for Cloud:** Need for Virtualization, Pros and cons of Virtualization, Types of virtualizations, System VM, Virtual Machine monitor, Virtual Machine properties, Interpretation and binary translation, HLL VM, Hypervisors, Xen, KVM, VMWare, Live VM migration steps and performance effects.	09
III	**Cloud Platform Architecture over Virtualized Data Centers:** Cloud Computing service Models: Public, private and Hybrid clouds, Data Center Design and Interconnection networks, Architectural Design of Compute and storage clouds, Public Cloud Platforms: GAE, AWS and Azure, Inter-cloud Resource Management, Cloud Security and Trust Management	08

IV	**Inter-Cloud Resource Management:** Extended Cloud Computing services, Resource provisioning and platform deployment, Virtual Machine creation and Management, Global Exchange of Cloud resources. **Cloud Security and Trust Management:** Cloud security defense strategies, Distributed intrusion/ anomaly detection, data and software protection techniques, Reputation guided protection of data centers.	**08**
V	**Cloud Computing Software Security Fundamentals:** Relevant Cloud Security Design Principles, Secure Development Practices, Approaches to Cloud Software Requirements Engineering, Cloud Security Policy Implementation and Decomposition, NIST 33 Security Principles. **Secure Cloud Software Testing:** Testing for Security Quality Assurance, Cloud Penetration Testing, Regression Testing, General Principle and Practices, Outsourcing, CSA STAR.	**08**

Text Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	Cloud Computing: A Hands- on Approach	Arshdeep Bagha, Vijay Madisetti	1st Edition, University Press, 2014
2	Cloud Security- A Comprehensive Guide to secure cloud computing	Ronald L. Krutz and Russell Dean Vines	1st Edition, Wiley, 2010
3	Distributed and cloud Computing from Parallel Processing to the Internet of Things	Kai Hwang, Geoffrey C. Fox, Jack J Dongarra	1st Edition, Morgan Kaufmann, Elsevier, 2013

Reference Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	Virtual Machines	James E Smith, Ravi Nair	1st Edition, Morgan Kaufmann, Publishers, 2006
2	Cloud Computing – Principles and Paradigms	Rajkumar Buyya, Broberg and rzejGoscinski	1st Edition, WILEY: A JOHN WILEY & SONS, INC., 2011

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs150
https://onlinecourses.nptel.ac.in/noc26_cs150/preview

Course Outcomes:

Course outcome	Descriptions
CO1	Analyze and interpret the concepts of Neural Networks.
CO2	Apply Deep Learning processes and their statistical properties to train Deep models.
CO3	Analyze sequential data to build recurrent and recursive models using regularization and convolutional operations.
CO4	Develop Deep Learning models to solve real world problems
CO5	Develop and evaluate recurrent neural network (RNN) models and advanced sequence models like LSTM and encoder-decoder architectures for sequential data processing tasks.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3											2		
CO2		3												
CO3	3											2		
CO4			3		3				1	1				3
CO5														

Course Assessment and Evaluation: Integrated Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (m in)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory - Internal Assessment	IA-1	90	50	15	12
		IA-2	90	50	15	
2	Periodic Lab Assessment		-	30	10	4
3	Lab CIE		-	20	5	2
4	One Open Ended Experiment /Project		-	10	5	2
	Total CIE			**150**	**50**	**20**

IA-1 Syllabus: 2½ Module , IA-2 Syllabus: 2½ Module
Rubrics for Periodic Lab Assessment: Attendance (5M), Lab record (10M), Conduction of Experiments(10M) and Viva(5M)-Total 30M
Lab CIE: Write-up(5M), Conduction/Execution/Demonstration(10M) and Viva(5M)-Total 20M
One Open Ended Experiment /Project: Experimental/project Design (4 marks), Execution/demonstration (3 marks) and Analysis/Viva (3marks) – **Total 10M**

Semester End Examination (SEE): **Max 100 Marks,** Duration: 3 Hrs The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice. Each question may have a maximum of 4 sub-divisions.
Minimum passing Marks: 40% of Max Marks

Department: Department: Computer Science and Engineering (Cyber Security)		Semester:	VII
Subject: Digital Forensic and Incident Response			
Subject Code:	22CY702	L – T – P - S:	42-0-28-50 (In Hrs)
		Credits:	4

Sl. No	Course Objectives
1	Understand the principles and processes of cyber forensics and the role of digital evidence in investigations
2	Gain expertise in operating system forensics, including Windows, Linux, and Mac OS.
3	Explore specialized forensic domains like mobile devices, cloud environments, and malware.
4	Investigate forensic techniques for web attacks, emails, and solid-state devices.
5	Learn legal and technical aspects of cyber forensics, including Bitcoin forensics and report preparation.

Note: Use standard Keywords (Verbs) as in Blooms Level and bold the keywords

Unit	Description	Hrs
I	**Understanding the Digital Forensics Profession and Investigations:** An Overview of Digital Forensics, Preparing for Digital Investigations, Maintaining Professional Conduct, Preparing Digital Forensics Investigations, Procedures for Private Sector High Tech Investigations, Understanding Data Recovery Workstations and Software, Conducting an Investigation.	09
II	**Working with Windows and CLI Systems**: Understanding File Systems, Exploring Microsoft File Structures, Examining NTFS Disks, Understanding Whole Disk Encryption, Understanding the Windows Registry, Understanding Microsoft Startup Tasks, Understanding Virtual Machines.	09

III	**Data Acquisition:** Understanding Storage Formats for Digital Evidence, Determining the best Acquisition Method, Contingency Planning for Image Acquisitions, Using Acquisition Tools, Validating Data Acquisitions, Performing RAID acquisitions, Using Remote Network Acquisition Tools, other acquisition Tools.	**08**
IV	**Processing Crime and Incident Scenes:** Identifying Digital Evidence, Collecting Evidence in Private Sector Incidence Scenes, preparing for a Search, Securing Digital Incidence, Seizing Digital Evidence at the Scene, Obtaining Digital Hash and Reviewing the Case. Computer Forensics Analysis and Validation: Determining What Data to Collect and Analyze, Validating Forensic Data, Addressing Data-Hiding Techniques.	**08**
V	**Virtual Machines Forensics and Network Forensics:** An overview of Virtual Machines Forensics, Performing Live Acquisitions, Network Forensic Overview, Developing Procedures for Network Forensics, E-mail Investigations: Exploring the Role of E-mail in Investigations, Exploring the Roles of the Client and Server in E-mail, Investigating E-mail Crimes and Violations, Understanding E-mail Servers, Using Specialized E-mail Forensics Tools.	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Guide to Computer Forensics and Investigations	Bill Nelson, Amelia Phillips, Chris Steuart	Cengage Learning, 6th Edition, 2019.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Digital Forensics and Incidence Response	Gerard Johansen	3rd Edition Packt Publications. 2022.
2	Computer Forensics and Digital Investigation Forensic e-book	Suzanne Widup	3rd Edition, McGraw-Hill Publication.

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc26_cs143/preview
https://onlinecourses.nptel.ac.in/noc26_cs142/preview

Course Outcomes:

Course outcome	Descriptions
CO1	Understand basic forensic procedures and analysis planning, and identify common tools used in digital forensic analysis.
CO2	Describe file systems and booting procedures.
CO3	Illustrate how to find deleted or obfuscated file system objects and track past computer and Internet activity, including establishing timelines.
CO4	Identify techniques for embedding covert information in text, document, and image files.
CO5	Demonstrate knowledge of forensic tools.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2			2							1	2
CO2	1	2		2								2	
CO3	2	2	2		1								2
CO4		2	2	3	2								2
CO5	1	2			3					1			2

Course Assessment and Evaluation: Integrated Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl No	**Components**		**Duration(min)**	**Marks Conducted for**	**Marks Reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	15	12
		IA-2	90	50	15	
2	Periodic Lab Assessment		-	30	10	4
3	Lab CIE		-	20	5	4
4	One Open Ended Experiment /Project		-	10	5	2
	Total CIE			**150**	**50**	**20**

IA-1 Syllabus: 2½ Module , IA-2 Syllabus: 2½ Module
Rubrics for Periodic Lab Assessment: Attendance (5M), Lab record (10 M), Conduction of Experiments(10M) and Viva(5M)-**Total 30M**
Lab CIE: Write-up(5M), Conduction/Execution/Demonstration(10M) and Viva(5M)-**Total 20M**
One Open Ended Experiment /Project: Experimental/project Design (4 marks), Execution/demonstration (3 marks) and Analysis/Viva (3marks) – **Total 10M**

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice. Each question may have a maximum of 4 sub-divisions.
Minimum passing Marks: 40% of Max Marks

Department: CSE (Cyber Security)			Semester:	VII
Subject: Data Mining and Machine Learning				
Subject Code:	22CY7PE31		L – T – P - S:	42 – 0 – 0 – 48 (In Hrs)
			Credits	4

Sl. No	Course Objectives
1	Learn Data Mining concepts, preprocessing techniques
2	Understand associate rules for frequent pattern mining and learn concept of supervised and unsupervised learning techniques
3	Discuss various classification algorithms.
4	Be familiar with basic machine learning algorithms with classification and clustering.
5	Gain experience of doing in dependent study and research

Unit	Description	Hrs
I	**Introduction To Data Mining:** Introduction, what is Data Mining, Definition, Knowledge Discovery from Data (KDD), What Kinds of Data can be Mined, Data Mining Tasks, Integration of Data Mining System with a Database or Data Warehouse System, Types of Data Sets and Attribute Values. **Preprocessing:** Data Quality, Major Tasks in Data Pre-processing, Data Cleaning and Data Integration, Data Reduction, Data Transformation and Data Discretization, Measures of Similarity and Dissimilarity-Basics	9
II	**Introduction To Machine Learning:** Introduction to machine learning, learning problems and scenarios, need for machine learning, types of learning, standard learning tasks **Association Rules:** Problem Definition, Frequent Item Set Generation, The APRIORI Principle, Support and Confidence Measures, Association Rule Generation; APRIOIRI Algorithm, Compact Representation of Frequent Item Set- Maximal Frequent Item Set.	9

III	**Supervised Learning**: Learning a Class from Examples, Linear, Non-linear, Multi-class and Multi-label classification, Perceptron: multilayer neural networks back propagation - learning neural networks structures support vector machines: soft margin SVM going beyond linearity generalization and over fitting regularization validation. Decision Trees: ID3, Classification and Regression Trees (CART), Regression: Linear Regression, Multiple Linear Regression, Logistic Regression.	**8**
IV	**Unsupervised Learning:** Introduction to clustering, Hierarchical: AGNES, DIANA, Partitional: K-means clustering, K-Mode Clustering, Self-Organizing Map, Expectation Maximization, Gaussian Mixture Models, Principal Component Analysis (PCA), Locally Linear Embedding (LLE), Factor Analysis, Neural Networks: Introduction, Perceptron, Multilayer Perceptron, Support vector machines: Linear and Non Linear, Kernel Functions, K-Nearest Neighbors.	**8**
V	**Ensemble And Probabilistic Learning:** Ensemble Learning Model Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Gradient Boosting, Xg boost, Stacking, Bayesian Learning, Bayes Optimal Classifier, Naïve Bayes Classifier, Bayesi an Belief Networks, Mining Frequent Patterns	**8**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Data Mining: Concepts and Techniques	Jiawei Han, Micheline Kamber	Jian Pei (2012), 3rd edition, Elsevier, United States of America.
2	Machine Learning Algorithms	Giuseppe Bonaccorso	2nd Edition, Packt, 2018.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Data Mining Techniques	Arun K Pujari	3rd Edition, Universities Press.
2	Introduction to Machine Learning	Ethem Al Paydin	PHI ,2004

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc26_cs74/preview
https://onlinecourses.nptel.ac.in/noc26_cs77/preview
https://nptel.ac.in/courses/106105174

Course Outcomes:

Course outcome	Descriptions
CO1	Perform the preprocessing of data and apply mining techniques on it
CO2	Recognize the characteristics of machine learning that makes it useful to solve real-world problems
CO3	Design and implement machine learning solutions of classification, regression problems.
CO4	Choose an appropriate clustering technique to solve real world problems.
CO5	Choose a suitable machine learning model, implement and examine the performance of the chosen model for a given real-world problem.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1								1					3
CO2					2	1		1					2
CO3	1	2	3	1	3							2	
CO4	2	1	3	1	2	1						2	
CO5	1	2	1	1	1	1						1	

Course Assessment and Evaluation: Regular Theory Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration(min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	AB L1	-	25	5	4
		AB L2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module, IA-2 Syllabus: 2 ½ Module
Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and Company specific quiz), Student Development program (1 week/online or offline), ……Etc
Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring **100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**. **Minimum passing Marks: 40% of Max Marks**

Department: Department: Computer Science and Engineering (Cyber Security)		Semester:	VII
Subject: Cyber Threat Intelligence			
Subject Code:	**22CY7PE32**	**L – T – P - S:**	**42-0-0-48 (In Hrs)**
		Credits:	**3**

Sl. No	Course Objectives
1	Comprehend the fundamentals of Blockchain and its organization.
2	Describe the underlying concepts of working of a Blockchain.
3	Infer the working principle of Bitcoin.
4	Interpret the working of Blockchain using Ethereum.
5	Examine possible business applications of Blockchain.

Unit	Description	Hrs
I	**Introduction to Threat Intelligence:** What is Threat Intelligence? What have you heard about threat Intelligence? Why is Threat Intelligence is important? Who can benefit from Threat Intelligence, Data and information are not intelligence, Two types of Threat Intelligence, The Role of Threat Data Feeds, The Role of Private Channels and the Dark Web? Threat Intelligence Lifecycle: The Six Phases of the Threat, Intelligence Lifecycle, Tools and People.	**09**
II	**Threat Intelligence for Security Operations:** Responsibilities of the SOC Team, The Overwhelming Volume of Alerts, Use case: Correlating and enriching alerts. Threat Intelligence for Incident response: Continuing Challenges, The Reactivity Problem, Minimizing Reactivity in Incident Response, Strengthening Incident Response with Threat, Intelligence, Threat Intelligence in Action, Essential Characteristics of Threat Intelligence for Incident Response. Threat Intelligence for Vulnerability Management: The Vulnerability Problem by the Numbers, Assess Risk Based on Exploitability.	**09**

III	**The Genesis of Threat Intelligence**: Vulnerability Databases, Threat Intelligence and Real Risk, Sources of Intelligence, Use Case: Cross-Referencing Intelligence. Threat Intelligence for Security Leaders: Risk Management, Mitigation, Investment, Communication, Supporting Security Leaders, The Security Skills Gap, Intelligence to Manage Better. Threat Intelligence for Risk Analysis: A Fair Risk Model, Threat Intelligence for Fraud Prevention: Stand and Deliver, Know Your Enemy, Criminal Communities and the Dark Web, Connecting the Dots for Fraud Prevention. Use cases: Payment fraud, Compromised data, Typo-squatting and fraudulent domains.	**08**
IV	**Threat Intelligence for Reducing Third Party Risk:** Third-Party Risk Looms Large, Traditional Risk Assessments Fall Short, Three Things to Look for in Threat Intelligence, Responding to High Third-Party Risk Scores. Threat intelligence for Digital Risk Protection: Being Online ids Being at Risk, Types of Digital Risk, Uncovering Evidence of breaches on the Web. Analytical Frameworks for Threat Intelligence: The Lockheed Martin Cyber Kill Chain, The Diamond Model, The MITRE ATT & CK Framework.	**08**
V	**Threat Intelligence journey:** Threat Intelligence Needs and Goals, Key Success Factors, Start Simple and Scale Up. Developing the core threat intelligence Team: Dedicated teams, Core Competencies, Collecting and Enriching Threat Data, Engaging with Threat Intelligence Communities, Conclusion: Moving Toward a Security Intelligence Program.	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	The Threat Intelligence Handbook Second edition	Christoper Ahlberg	Zane Pokorny CyberEdge Press, 2019.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Cyber threat intelligence	Dehghantanha , Ali, Mauro Conti, and Tooska Dargahi	New York, NY: Springer International Publishing, 2018.
2	Intelligence-driven incident response: Outwitting the adversary,	Roberts, Scott J., And Rebekah Brown	O'Reilly Media, 2017.

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc23_cs127/preview
https://onlinecourses.nptel.ac.in/noc26_cs142/preview
https://onlinecourses.nptel.ac.in/noc26_cs150/preview

Course Outcomes:

Course outcome	Descriptions
CO1	Understand the genesis and fundamentals of cyber threat intelligence, and process of managing cyber threats.
CO2	Explain vulnerability management and security incident response process.
CO3	Perform security risks analysis using various risk analysis tools.
CO4	Apply threat intelligence to assess and manage internal and third-party security risks.
CO5	Apply the principles of threat intelligence to define organizational needs and goals

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	1									1	2
CO2	2	1											2
CO3	1	2			1								2
CO4	2	1	2		2								2
CO5	3	2											2

Course Assessment and Evaluation: Regular Theory Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	AB L1	-	25	5	4
		AB L2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module
Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline), ……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs
The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice. Each question may have a maximum of 4 sub-divisions. Minimum passing Marks: 40% of Max Marks

Department: CSE (Cyber Security)			Semester:	VII
Subject: Web Application Security				
Subject Code:	22CY7PE33		L – T – P - S:	42 – 0 – 0 - 48
			Credits	3

Sl. No	Course Objectives
1	To understand the fundamentals of web application security.
2	To focus on wide aspects of secure development and deployment of web applications
3	To learn how to build secure APIs
4	To learn the basics of vulnerability assessment and penetration testing
5	To get an insight about Hacking techniques and Tools

Unit	Description	Hrs
I	**Fundamentals of Web Application Security:** The history of Software Security-Recognizing Web Application Security Threats, Web Application Security, Authentication and Authorization, Secure Socket layer, Transport layer Security, Session Management-Input Validation	8
II	**Secure Development and Deployment:** Web Applications Security - Security Testing, Security Incident Response Planning, The Microsoft Security Development Lifecycle (SDL), OWASP Comprehensive Lightweight Application Security Process (CLASP), The Software Assurance Maturity Model (SAMM)	8
III	**Secure API Development:** API Security- Session Cookies, Token Based Authentication, Securing Natter APIs: Addressing threats with Security Controls, Rate Limiting for Availability, Encryption, Audit logging, securing service-to-service APIs: API Keys, OAuth2, Securing Microservice APIs: Service Mesh, Locking Down Network Connections, Securing Incoming Requests.	9

IV	**Vulnerability Assessment and Penetration Testing:** Vulnerability Assessment Lifecycle, Vulnerability Assessment Tools: Cloud-based vulnerability scanners, Host-based vulnerability scanners, Network-based vulnerability scanners, Database based vulnerability scanners, Types of Penetration Tests: External Testing, Web Application Testing, Internal Penetration Testing, SSID or Wireless Testing, Mobile Application Testing.	**9**
V	**Hacking Techniques and Tools:** Social Engineering, Injection, Cross-Site Scripting (XSS), Broken Authentication and Session Management, Cross-Site Request Forgery, Security Misconfiguration, Insecure Cryptographic Storage, Failure to Restrict URL Access, Tools: Comodo, OpenVAS, Nexpose, Nikto, Burp Suite, etc.	**8**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Web Application Security	Andrew Hoffman	First Edition, 2020, O'Reilly Media, Inc.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Web Application Security: A Beginners Guide	Bryan Sullivan, Vincent Liu	2012, The McGraw-Hill Companies.
2	API Security in Action	Neil Madden	2020, Manning Publications Co., NY, USA.

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc25_cs142/preview
https://nptel.ac.in/courses/106106129?utm_source=chatgpt.com
https://nptel.ac.in/courses/106106199?utm_source=chatgpt.com
https://nptel.ac.in/courses/128106006?utm_source=chatgpt.com

Course Outcomes:

Course outcome	Descriptions
CO1	Understanding the basic concepts of web application security and the need for it
CO2	Be acquainted with the process for secure development and deployment of web applications
CO3	Acquire the skill to design and develop Secure Web Applications that use Secure APIs
CO4	Be able to get the importance of carrying out vulnerability assessment and penetration testing
CO5	Acquire the skill to think like a hacker and to use hackers tool sets

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1								1						3
CO2					2	1		1						2
CO3	1	2	3	1	3								2	
CO4	2	1	3	1	2	1							2	
CO5	1	2	1	1	1	1							1	

Course Assessment and Evaluation: Regular Theory Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline), ……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice. Each question may have a maximum of 4 sub-divisions.
Minimum passing Marks: 40% of Max Marks

Department: Department: Computer Science and Engineering (Cyber Security)			Semester:	VII
Subject: Information Retrieval for Network Security				
Subject Code:	22CY7PE41	L – T – P - S:	42-0-0-48 (In Hrs)	
		Credits:	3	

Sl. No	Course Objectives
1	Learn fundamental concepts of information retrieval and indexing techniques for security data
2	Understand various information retrieval models and ranking mechanisms applicable to security contexts
3	Understand network security fundamentals and threat landscape
4	Apply information retrieval techniques to detect and analyze security threats and anomalies
5	Implement advanced security analytics solutions using IR and machine learning techniques

Note: Use standard Keywords (Verbs) as in Blooms Level and bold the keywords

Unit	Description	Hrs
I	**Fundamentals of Information Retrieval:** Introduction to Information Retrieval, Definition and scope of IR, Text Processing and Analysis, Tokenization, Stemming, Lemmatization, Stop Word Removal. Indexing: Inverted Index, B-tree Indexing, Hash-based Indexing, Index Organization, Index Compression Techniques, Index Maintenance.	**09**
II	**Information Retrieval Models:** Boolean Model, Vector Space Model, Probabilistic Models, Language Models, Query Formulation, Query Expansion, Query Refinement and Relevance Feedback, Ranking and Similarity Measures (Cosine Similarity, TF-IDF, BM25).	**08**

III	**Network Security Fundamentals:** Overview of Network Security, Threats and Vulnerabilities, Cryptography Basics (Symmetric and Asymmetric Encryption), Digital Signatures, Certificates and PKI, Security Protocols (SSL/TLS, IPSec), Intrusion Detection Systems (IDS), Firewalls and Proxies, Network Sniffing Detection.	**08**
IV	**Information Retrieval in Security Analytics:** Log Analysis and Parsing, Event Correlation and Aggregation, Anomaly Detection using Information Retrieval Techniques, Malware Detection through Pattern Matching, Threat Intelligence Integration, Security Information and Event Management (SIEM)Systems, Big Data Analytics for Security.	**09**
V	**Advanced Applications and Emerging Trends:** Web Security and Search Engine Optimization, Privacy Preserving Information Retrieval, Semantic Web and Ontologies for Security, Machine Learning in Information Retrieval for Security, Deep Learning Approaches for Cyber Threat Detection, Case Studies in Network Security Information Systems, Ethical and Legal Aspects of Security Data Mining	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Introduction to Information Retrieval	Christopher D. Manning, Prabhakar Raghavan, Hinrich Schütze	Cambridge University Press,2008
2	Cryptography and Network Security: Principles and Practice	William Stallings	7th Edition, Pearson, 2017
3	Mining of Massive Datasets	Jure Leskovec, Anand Rajaraman, Jeffrey D. Ullman	Cambridge University Press,3rd Edition, 2020

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Modern Information Retrieval	Ricardo Baeza-Yates, Berthier Ribeiro-Neto	Addison-Wesley, 2ndEdition, 2011
2	Network Security Essentials	William Stallings	Pearson, 7th Edition,2017
3	SIEM Implementation: Building Successful Security Information and Event Management Systems	Merike Kaeo Elsevier	Elsevier, 2013
4	Cyber Security Analytics	Marshall, Reed Sage, & Christopher	Syngress, 2014

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc25_ee54/preview?utm_source=chatgpt.com
https://nptel.ac.in/courses/106106248?utm_source=chatgpt.com
https://nptel.ac.in/courses/106106129?utm_source=chatgpt.com
https://nptel.ac.in/courses/106105162

Course Outcomes:

Course outcome	Descriptions
CO1	Apply text preprocessing and indexing techniques to organize and retrieve security-related information efficiently
CO2	Evaluate and select appropriate information retrieval models for network security applications
CO3	Understand network security threats, vulnerabilities, and protection mechanisms
CO4	Implement anomaly detection and threat analysis using information retrieval and data mining techniques
CO5	Develop integrated security analytics solutions combining information retrieval with machine learning for cyber threat detection and response.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	3										3	3
CO2	2	2	1									3	2
CO3	1	2	3	2								3	3
CO4	2	3	2	1	1							3	3
CO5	2	3	3	1	2	1	1				1	3	3

Course Assessment and Evaluation: Regular Theory Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl.No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½
Module Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1week/online or offline),……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs
The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice.Each question may have a maximum of 4 sub-divisions. Minimum passing Marks: 40% of Max Marks

Department: Computer Science and Engineering (Cyber Security)			Semester:	VII
Subject: Malware Analysis and Reverse Engineering				
Subject Code:	22CY7PE42	L – T – P - S:	42-0-0-48 (In Hrs)	
		Credits:	3	

Sl. No	Course Objectives
1	To understand the fundamentals of malware, its characteristics, and classification
2	To learn static and dynamic malware analysis techniques and methodologies
3	To understand reverse engineering concepts and assembly language programming
4	To learn advanced reverse engineering techniques for handling obfuscated and anti-analysis malware

Unit	Description	Hrs
I	**Fundamentals of Malware:** Introduction to Malware, Definition and Classification of Malware, Types of Malwares (Viruses, Worms, Trojans, Rootkits, Ransomware, Spyware, Adware, Botnets), Malware Characteristics and Behavior, Malware Distribution Mechanisms, Malware Evolution and Variants, Malware Propagation Techniques, Common Attack Vectors.	09
II	**Static and Dynamic Malware Analysis:** Static Analysis Techniques: File Format Analysis, Disassembly Fundamentals, Code Structure Analysis, Metadata Extraction, Packing Detection and Unpacking. Dynamic Analysis Techniques: Behavioral Analysis, Sandboxing, System Call Monitoring, Network Traffic Analysis, Registry and File System Monitoring, Debugging Techniques, Trace Analysis.	08

III	**Reverse Engineering Fundamentals:** Introduction to Reverse Engineering, Assembly Language Basics (x86 and x64 Architecture), Instruction Sets and Opcodes, Control Flow Analysis, Function Recognition, Loop and Conditional Structure Analysis, API Identification, Import Address Table (IAT) Analysis, De compilation Techniques.	**08**
IV	**Advanced Reverse Engineering:** Binary Obfuscation Techniques and De-obfuscation, Anti-Analysis Techniques (Anti-debugging, Anti-disassembly, Anti-VM, Anti-sandboxing), Circumventing Anti-Analysis Mechanisms, Cryptanalysis for Malware, Malware Configuration Extraction, Strings and Resource Analysis, Advanced Debugging with Tools.	**09**
V	**Malware Analysis Tools and Case Studies:** Malware Analysis Tools: IDA Pro, Ghidra, OllyDbg,WinDbg, Wireshark, Process Monitor, Regshot, VirusTotal, Cuckoo Sandbox. Practical Case Studies: Real-world Malware Analysis Examples (Stuxnet, WannaCry, Mirai, Zeus), Threat Intelligence Integration, Malware Report Writing and Documentation, Ethical and Legal Considerations in Malware Analysis.	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software	Michael Sikorski, Andrew Honig	No Starch Press, 1st Edition, 2012
2	The IDA Pro Book: The Unofficial Guide to the World's Most Popular Disassembler	Chris Eagle	No Starch Press, 2nd Edition, 2011
3	Reverse Engineering: Secrets of Reverse Engineering	Eldad Eilam	Wiley Publishing, 2005

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Modern Malware Forensics: Fighting Malicious Code	Cameron H. Malin, Eoghan Casey, James M. Aquilina	Syngress, 1st Edition, 2013
2	Malware and Rootkits Forensics: Analyzing and Reverse-Engineering Malicious Code	Anatoli Arkhipenko	Syngress, 2014
3	Assembly Language Step-by-Step: Programming with Linux	Jeff Duntemann	Wiley, 3rd Edition,2009
4	Penetration Testing and Network Defense	Georgia Weidman	Syngress, 2015

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc25_cs142/preview

https://onlinecourses.nptel.ac.in/noc23_cs127/preview

https://nptel.ac.in/courses/106106129?utm_source=chatgpt.com

https://onlinecourses.nptel.ac.in/noc25_ee54/preview?utm_source=chatgpt.com

Course Outcomes:

Course outcome	Descriptions
CO1	Understand various types of malwares, their characteristics, propagation mechanisms, and evolution
CO2	Perform static and dynamic analysis of malware to identify behaviours and extract indicators of compromise
CO3	Apply reverse engineering techniques to understand binary code structure and program functionality.
CO4	Analyze and overcome anti-analysis and obfuscation techniques employed by advanced malware.
CO5	Use industry-standard tools to analyze real-world malware samples and generate comprehensive threat intelligence reports.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2										3	3
CO2	2	3	1	3	1							3	3
CO3	3	3	2	2	2							3	3
CO4	3	3	2	3	1			1			1	3	3
CO5	2	3	3	2	3	1				1	1	3	3

Course Assessment and Evaluation: Regular Theory Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Mark s reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module
Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline),……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs
The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus **coverage.**
Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**.
Minimum passing Marks: 40% of Max Marks

Department: CSE (Cyber Security)			Semester:	VII
Subject: Distributed Systems				
Subject Code:	22CY7PE43		L – T – P – S:	42-0-0-48 (In Hrs)
			Credits	3

Unit	Description	Hrs
I	**Introduction**: Definition–Relation to computer system components –Motivation – Relation to parallel systems – Message-passing systems versus shared memory systems – Primitives for distributed communication Synchronous versus asynchronous executions A model of distributed computations: A distributed program –A model of distributed executions –Models of communication networks –Global state – Cuts –Past and future cones of an event –Models of process communications. Logical Time: A framework for a system of logical clocks –Scalar time –Vector time – Physical clock synchronization: NTP	09
II	**Message Ordering & Snapshots** **Message ordering and group communication:** Message ordering paradigms – Asynchronous execution with synchronous communication Synchronous program order on an asynchronous system –Group communication – Causal order (CO) - Total order. **Global state and snapshot recording algorithms:** Introduction –System model and definitions –Snapshot algorithms for FIFO channels	09
III	**Distributed Mutex & Deadlock** **Distributed mutual exclusion algorithms:** Introduction – Preliminaries – Lamport‘s algorithm – Ricart - Agrawala algorithm – Suzuki – Kasami‘s broadcast algorithm. **Deadlock detection in distributed systems:** Introduction – System mode – Preliminaries –Models of deadlocks – Knapp‘s classification of Distributed Deadlock detection algorithm–	08

	Mitchell and merits Algorithms for the single resource model.	
IV	**Recovery & consensus** **Check pointing and rollback recovery:** Introduction – Background and definitions – Issues in failure recovery – Checkpoint-based recovery – Log-based rollback recovery – Koo-toueg Coordinated checkpointing algorithm – Juang-Venkateshar Algorithm for asynchronous checkpointing and recovery. **Consensus and agreement algorithms:** Problem definition – Overview of results – Agreement in a failure – free system – Agreement in synchronous systems with failures.	**08**
V	**P2P & distributed shared memory** **Peer-to-peer computing and overlay graphs:** Introduction – Data indexing and overlays – Chord distributed hash table – Content addressable networks – Tapestry. **Distributed shared memory:** Abstraction and advantages – Memory consistency models –Shared memory Mutual Exclusion.	**08**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Distributed computing: principles, algorithms, and systems	Kshemkalyani, Ajay D., and Mukesh Singhal	First Edition, Cambridge University Press, 2011

Reference Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	Distributed Systems Concepts and Design	George Coulouris, Jean Dollimore and Tim Kindberg,	5th Edition, Pearson Education, 2012
2	Distributed Systems: Principles and Paradigms	Tanenbaum A.S., Van Steen M.	Second Edition, Pearson Education, 2007

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc26_cs29/preview?utm_source=chatgpt.com
https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs115
https://onlinecourses.nptel.ac.in/noc25_cs84/preview?utm_source=chatgpt.com

Course Outcomes:

Course outcome	Descriptions
CO1	Apply knowledge of Mutual Exclusion, Deadlock detection algorithms in distributed systems
CO2	Analyze the issues of distributed systems, synchronization issues and global state for distributed systems
CO3	Apply agreement protocols and fault tolerance mechanisms in distributed systems for peer-to-peer and distributed shared memory systems
CO4	Analyze checkpointing, rollback recovery, and consensus algorithms to ensure fault tolerance and agreement in distributed systems
CO5	Evaluate peer-to-peer computing, distributed shared memory architectures, and memory consistency models for designing scalable distributed systems.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PO3
CO1	3											2		
CO2		3										2		
CO3	3											2		
CO4		3												
CO5			3											

Course Assessment and Evaluation: Regular Theory Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration(min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline), ……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs
The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus **coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**. **Minimum passing Marks: 40% of Max Marks**

Department: CSE (Cyber Security)			Semester:	VII
Subject: Introduction to Deep Learning				
Subject Code:	22CY7PE51		L – T – P - S	42-0-0-50
			Credits	3

	Course Objectives
1	Understand the fundamentals of deep learning.
2	Understand the working of Convolutional Neural Networks and Recurrent Neural Network in decision making.
3	Learn the strength and weaknesses of deep learning approaches.
4	Explore the applications of deep learning to solve real world problems.

Unit	Description	Hrs
I	**Introduction:** What is a Neural Network?, The Human Brain, Models of a Neuron, Neural Networks Viewed As Directed Graphs, Feedback, Network Architectures, Rosenblatt‟s Perceptron: Introduction, Perceptron, The Perceptron Convergence Theorem, Relation Between the Perceptron and Bayes Classifier for a Gaussian Environment.	09
II	**Multilayer Perceptron:** Introduction, Batch Learning and On-Line Learning, The Back-Propagation Algorithm, XOR Problem, Heuristics for Making the Back- Propagation Algorithm Perform Better, Back Propagation and Differentiation.	09

III	**Regularization for Deep Learning:** Parameter Norm Penalties - L2 Parameter Regularization, Dataset Augmentation, Semi-Supervised Learning. Optimization for Training Deep Models: Challenges in Neural Network Optimization – Ill Conditioning, Local Minima, Plateaus, Saddle Points and Other Flat Regions.	**08**
IV	**Convolution Neural Networks:** The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data Types, Efficient Convolution Algorithms, Convolutional Networks and the History of Deep Learning.	**08**
V	**Sequence Modeling**: Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to- Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks, The Long Short-Term Memory and Other Gated RNNs	**08**

Text Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	Neural networks and Learning Machines	Simon Haykin	Third Edition, Third Edition,2016
2	Deep Learning	Ian Goodfellow, Yoshua Bengio and Aaron Courville	MIT Press, 2016. https://www.deeplearning bo ok.org/lecture_slides.ht m l

Reference Books:

Sl. No	Text Book title	Author	Volume and Year of Edition
1	Machine Learning	Tom Mitchell	McGraw Hill, 3rd Edition, 1997.
2	Machine Learning - An Algorithmic Perspective	Stephen Marsland	Second Edition, CRC, Press - Taylor and Francis Group, 2015.

Web links and Video Lectures (e-Resources):
https://www.youtube.com/watch?v=VyWAvY2CF9c
https://www.youtube.com/watch?v=7sB052Pz0sQ

https://www.youtube.com/watch?v=Mubj_fqiAv8
https://www.coursera.org/learn/neural-networks-deep-learning

Course Outcomes:

Course outcome	Descriptions
CO1	Analyze and interpret the concepts of Neural Networks.
CO2	Apply Deep Learning processes and their statistical properties to train Deep models.
CO3	Analyze sequential data to build recurrent and recursive models using regularization and convolutional operations.
CO4	Develop Deep Learning models to solve real world problems
CO5	Develop and evaluate recurrent neural network (RNN) models and advanced sequence models like LSTM and encoder-decoder architectures for sequential data processing tasks.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2			2					2			
CO2	2	2	1										
CO3	2									2			
CO4			2										
CO5		3	3	2									

Course Assessment and Evaluation: Regular Theory Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline), ……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice.
Each question may have a maximum of 4 sub-divisions. Minimum passing Marks: 40% of Max Marks

Department: Department: CSE (Cyber Security)			Semester:	VII
Subject: Digital and Mobile Forensic				
Subject Code:	22CY7PE52	L – T – P - S:	42-0-0-48 (In Hrs)	
		Credits:	3	

Sl. No	Course Objectives
1	To understand basic digital forensics and techniques.
2	To understand digital crime and investigation.
3	To understand how to be prepared for digital forensic readiness
4	To understand and use forensics tools for iOS devices
5	To understand and use forensics tools for Android devices.

Unit	Description	Hrs
I	**INTRODUCTION TO DIGITAL FORENSICS:** Forensic Science – Digital Forensics – Digital Evidence – The Digital Forensics Process –Introduction – The Identification Phase – The Collection Phase – The Examination Phase – The Analysis Phase – The Presentation Phase	09
II	**DIGITAL CRIME AND INVESTIGATION:** Digital Crime – Substantive Criminal Law – General Conditions – Offenses – Investigation Methods for Collecting Digital Evidence – International Cooperation to Collect Digital Evidence	08
III	**DIGITAL FORENSIC READINESS:** Introduction – Law Enforcement versus Enterprise Digital Forensic Readiness – Rationale for Digital Forensic Readiness – Frameworks, Standards and Methodologies – Enterprise Digital Forensic Readiness – Challenges in Digital Forensics	08
IV	**IOS FORENSICS:** Mobile Hardware and Operating Systems - iOS Fundamentals – Jailbreaking – File System – Hardware – iPhone Security – iOS Forensics – Procedures and Processes –Tools – Oxygen Forensics – Mobil Edit – iCloud	08

V	**ANDROID FORENSICS:** Android basics – Key Codes – ADB – Rooting Android – Boot Process – File Systems – Security – Tools – Android Forensics – Forensic Procedures – ADB – Android Only Tools – Dual Use Tools–Oxygen Forensics – Mobil Edit – Android App Decompiling.	**09**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Digital Forensics	Andre Arnes	Wiley, 2018.
2	An In-depth Guide to Mobile Device Forensics	Chuck Easttom	First Edition, CRC Press, 2022.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Computer Forensics: Computer Crime Scene Investigation	John R. Vacca, Charles River Media.	2005, 2nd Edition
2	Guide to Computer Forensics and Investigations	Nelson, B., Phillips, A., & Steuart, C	2006, 2nd Edition

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc26_cs144/preview
https://onlinecourses.nptel.ac.in/noc25_cs142/preview
https://onlinecourses.nptel.ac.in/noc26_lw01/preview?utm_source=chatgpt.com
https://onlinecourses.swayam2.ac.in/cec20_lb06/preview?utm_source=chatgpt.com

Course Outcome

Course outcome	Descriptions
CO1	Interpret the basic concepts of pointers, structures and stacks.
CO2	Know about digital crime and investigations.
CO3	Be forensic ready
CO4	Investigate, identify and extract digital evidence from iOS devices
CO5	Investigate, identify and extract digital evidence from Android device

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	3	2	1	-	-	-	1	1	3		
CO2	3	3	3	3	3	-	-	-	2	2	1		
CO3	3	3	2	3	1	-	-	-	3	2	1		
CO4	3	1	2	2	3	-	-	-	1	3	3		
CO5	1	3	2	3	2	-	-	-	2	3	2		

Course Assessment and Evaluation: Regular Theory Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl.No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module
Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline), ……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice. Each question may have a maximum of 4 sub-divisions. Minimum passing Marks: 40% of Max Marks

Department: CSE (Cyber Security)			Semester:	VII
Subject: Cognitive IoT				
Subject Code:	22CY7PE53		L – T – P - S	42-0-0-48
			Credits	3

	Course Objectives
1	To understand the fundamentals of IoT architecture, devices, and communication protocols
2	To learn various IoT hardware platforms and their integration with cloud and edge computing
3	To understand data processing, analytics, and real-time stream processing in IoT systems
4	To apply machine learning and artificial intelligence techniques for intelligent IoT systems
5	To understand IoT security, privacy mechanisms, and implement cognitive IoT solutions in real-world applications

Unit	Description	Hrs
I	**Introduction to IoT and Cognitive Computing:** Overview of Internet of Things, IoT Architecture and Layers, IoT Devices and Sensors, IoT Communication Protocols (WiFi, Bluetooth, Zigbee, LoRaWAN, NB-IoT, 5G), Edge Computing and Fog Computing, Cloud Computing for IoT, Introduction to Cognitive Computing, Cognitive Systems Characteristics, Cognitive IoT Integration, Machine Learning in IoT.	08
II	**IoT Hardware and Platforms:** IoT Hardware Components: Microcontrollers, Microprocessors, Embedded Systems, Single-Board Computers (Raspberry Pi, Arduino, Beagle Bone), Actuators and Sensors, IoT Operating Systems (Tiny OS, RIOT, Contiki), IoT Platforms: AWS IoT, Azure IoT Hub, Google Cloud IoT, IBM Watson IoT, MQTT Protocol and Brokers, CoAP Protocol, Message Queuing and Data Integration.	08

III	**Data Processing and Analytics in Cognitive IoT:** Data Collection from IoT Devices, Data Preprocessing and Cleaning, Real-time Data Processing Frameworks (Apache Kafka, Apache Spark, Apache Flink), Time-Series Data Analysis, Stream Processing, Event Processing, Data Aggregation and Fusion, Analytics at Edge vs Cloud, Complex Event Processing (CEP), Big Data Technologies for IoT.	**08**
IV	**Machine Learning and Artificial Intelligence for IoT:** Machine Learning Algorithms for IoT (Regression, Classification, Clustering), Supervised and Unsupervised Learning, Feature Selection and Engineering, Model Training and Optimization, Predictive Analytics, Anomaly Detection in IoT Systems, Reinforcement Learning Applications, Deep Learning for IoT, Transfer Learning, Model Deployment on Edge Devices.	**09**
V	**IoT Security, Privacy, and Real-World Applications:** IoT Security Threats and Vulnerabilities, Secure Communication Protocols (DTLS, TLS), Authentication and Authorization in IoT, Encryption for IoT, Privacy-Preserving Techniques, Blockchain for IoT Security, Case Studies: Smart Cities, Smart Homes, Industrial IoT (IIoT), Healthcare IoT, Agriculture IoT (Precision Farming), Connected Vehicles, Environmental Monitoring, Cognitive IoT Applications and Future Trends.	**09**

Reference Books:

Building Internet of Things with Raspberry Pi and Arduino	Jayakumar Pawan Kumar, Kamesh Ramakrishnan	Packt Publishing, 2016
IoT Security Essentials: Design and Implementation	Francesca Bosco, Nicholas Pohlmann	Syngress, 2018
Deep Learning for Internet of Things	Arvind Kumar Pandey	Apress, 2021
Industrial Internet of Things: From the Ground Up	David Brolan, Sarah Frazelle	O'Reilly Media, 2018

Text Books:

Sl. No	Text Book Title	Author	Volume and Year of Edition
1	Internet of Things: A Hands-On Approach	Arshdeep Bahga, Vijay Madisetti	Universities Press, 1st Edition, 2014
2	Machine Learning for Internet of Things: A Comprehensive Survey	Dinesh C. Jain	Springer, 1st Edition, 2021
3	Cognitive IoT: Machine Learning for Smart Internet of Things	Sridhar Setlur, Pradeep Bandi	Packt Publishing, 2021

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc26_cs37/preview
https://onlinecourses.nptel.ac.in/noc26_cs132/preview
https://onlinecourses.nptel.ac.in/noc26_cs38/preview

Course Outcomes

Course outcome	Descriptions
CO1	Understand IoT architecture, hardware components, communication protocols, and platform integration
CO2	Design and implement IoT systems using edge computing, cloud platforms, and appropriate middleware solutions
CO3	Apply data processing and analytics techniques for real-time monitoring and analysis of IoT data streams
CO4	Implement machine learning and artificial intelligence algorithms to develop intelligent and predictive IoT applications
CO5	Design secure and privacy-aware cognitive IoT solutions and apply them to real-world domain-specific applications

Course Articulation Matrix

CO/ P O	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 1 0	PO 1 1	PS O1	PS O2
CO1	2	2	1	-	2	-	-	-	-	-	-	3	2
CO2	2	2	3	2	3	-	1	-	-	-	-	3	3
C03	2	3	2	3	3	-	-	-	-	-	-	3	3
C04	3	3	3	2	2	-	-	-	-	-	1	3	3
CO5	2	3	3	2	3	1	2	3		1	1	3	3

Course Assessment and Evaluation: Regular Theory Courses

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each compone nt to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline), ……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice.
Each question may have a maximum of 4 sub-divisions. Minimum passing Marks: 40% of Max Marks

Department: Department: Computer Science and Engineering (Cyber Security)		Semester:	VII
Subject: Cyber Forensics			
Subject Code:	22CY7PE61	L – T – P - S:	42-0-0-48 (In Hrs)
		Credits:	3

Sl. No	Course Objectives
1	Understand the principles and processes of cyber forensics and the role of digital evidence in investigations.
2	Gain expertise in operating system forensics, including Windows, Linux, and Mac OS.
3	Explore specialized forensic domains like mobile devices, cloud environments, and malware.
4	Investigate forensic techniques for web attacks, emails, and solid-state devices.

Unit	Description	Hrs
I	**Foundations of Cyber Forensics** Introduction to Cyber Forensics: Overview of cyber forensics principles and processes; Understanding the role of digital evidence in investigations. Windows Forensics: Analysis of Windows operating systems; Examining system logs, registry files, and memory dumps. Linux Forensics: Investigating Linux systems; Log analysis, file system structures, and identifying forensic artifacts.	08
II	**Advanced Operating System Forensics** Mac OS Forensics: Mac OS file systems and log analysis; Extracting artifacts and understanding unique challenges in Mac environments. Anti-forensics: Techniques used by attackers to cover their tracks; Methods to detect and counter anti-forensic measures.	08

	Network Forensics: Monitoring and analyzing network traffic for forensic evidence; Understanding packet captures, network logs, and intrusion detection systems.	
III	**Specialized Forensics Domains** Mobile Forensics: Techniques for extracting data from mobile devices; Investigating Android, iOS, and other mobile platforms. Cloud Forensics: Challenges in collecting evidence from cloud environments; Legal and technical considerations in cloud investigations. Malware Forensics: Analyzing malware samples to understand their behavior; Identifying and mitigating malware in forensic investigations.	**08**
IV	**Focused Attack and Device Forensics** Web Attack Forensics: Investigating web-based attacks, including SQL injection and cross site scripting; Identifying server logs and tracing attack vectors. Emails and Email Crime: Tracing email headers and identifying email-based attacks; Investigating phishing and fraud cases. Solid State Device (SSD) Forensics: Understanding unique challenges in SSD forensics; Techniques for recovering and analyzing data from SSDs.	**09**
V	**Legal and Emerging Domains** Bitcoin Forensics: Tracking cryptocurrency transactions and wallets; Investigating blockchain based crimes. Cyber Law and Cyberwarfare: Legal frameworks for handling digital evidence; Understanding international laws and cyberwarfare tactics. Investigative Reports and Legal Acceptance: Preparing forensic reports for legal proceedings; Ensuring the admissibility of evidence in court.	**09**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	The Threat Intelligence Handbook Second edition	Christoper Ahlberg	Zane Pokorny CyberEdge Press, 2019.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Cyber threat intelligence	Dehghantanha, Ali, Mauro Conti, and Tooska Dargahi	New York, NY: Springer International Publishing, 2018.
2	Intelligence-driven incident response: Outwitting the adversary,	Roberts, Scott J.,and Rebekah Brown	O'Reilly Media, 2017.

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc25_cs142/preview
https://onlinecourses.nptel.ac.in/noc25_cs120/preview
https://onlinecourses.nptel.ac.in/noc26_lw01/preview?utm_source=chatgpt.com

Course Outcomes:

Course outcome	Descriptions
CO1	Explain foundational concepts and processes in cyber forensics and digital evidence handling.
CO2	Analyze forensic artifacts across various operating systems and network environments.
CO3	Apply forensic methods to specialized domains like mobile devices, cloud environments, and malware analysis.
CO4	Investigate and trace evidence related to web attacks, email crimes, and SSDs.
CO5	Explain Bitcoin Forensics

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	1									1	2
CO2	2	1											2
CO3	1	2			1								2
CO4	2	1	2		2								2
CO5	3	2											2

Course Assessment and Evaluation: Regular Theory Courses

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Continuous Internal Evaluation (CIE): 50 Marks						
Sl.No	Components		Duration(min)	Marks Conducted for	Marks reduced to	40% Min Marks in each component to write SEE
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module
Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline), …Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice.

Department: CSE (Cyber Security)			Semester:	VII
Subject: Social Network Security				
Subject Code:	22CY7PE62	L – T – P -S:	42-0-0-48	
		Credits:	3	

Sl. No	Course Objectives
1	To **understand** the fundamentals of social networks, their structure, and analysis techniques
2	To **identify** and analyze security threats and vulnerabilities in online social networks
3	To **learn** privacy protection mechanisms and access control models for social media platforms
4	To **apply** security techniques to detect and prevent attacks on social networks

Unit	Description	Hrs
I	**MODULE 1: Introduction to Social Networks (8 Hours)** • Evolution and Growth of Social Networks • Types of Social Networks: Facebook, Twitter, LinkedIn, Instagram, TikTok • Social Network Structure and Components • Graph Theory Fundamentals: Nodes, Edges, Directed and Undirected Graphs • Network Metrics: Degree Centrality, Betweenness, Closeness, Eigenvector Centrality • Community Detection and Clustering • Social Network Analysis Tools and Techniques • Small World Phenomenon and Six Degrees of Separation	9
II	**MODULE 2: Security Threats in Social Networks (8 Hours)** • Threat Landscape in Online Social Networks • Privacy Violations and Information Leakage • Identity Theft and Profile Cloning	8

	• Social Engineering Attacks: Phishing, Spear Phishing, Whaling • Cyberstalking and Cyberbullying • Malware Propagation through Social Media • Cross-Site Scripting (XSS) and Cross-Site Request Forgery (CSRF) • Sybil Attacks and Fake Account Creation • Clickjacking and Like-jacking Attacks	
III	**MODULE 3: Privacy and Access Control (8 Hours)** • Privacy Models and Privacy Settings in Social Networks • Information Disclosure and Oversharing Risks • Privacy Paradox in Social Media • Access Control Models: DAC, MAC, RBAC, ABAC • Social Network Based Access Control (SNAC) • Relationship-Based Access Control (ReBAC) • Data Anonymization Techniques: k-anonymity, l-diversity, t-closeness • De-anonymization Attacks and Countermeasures • Privacy-Preserving Data Publishing	**8**
IV	**MODULE 4: Attack Detection and Prevention (8 Hours)** • Spam Detection in Social Networks • Machine Learning for Malicious Content Detection • Fake Profile and Bot Detection Techniques • Sentiment Analysis for Threat Detection • Phishing Detection using NLP and Deep Learning • Rumor and Misinformation Detection • Malicious URL Detection and Blacklisting • Social Network Honeypots • Intrusion Detection Systems for Social Media • CAPTCHA and Challenge-Response Mechanisms	**8**

V	**MODULE 5: Ethical, Legal and Emerging Trends in Social Network Security** Ethical Issues and Responsible Use of Social Media Data, Digital Footprints and User Consent, Legal and Regulatory Frameworks for Social Media Security and Privacy, Data Protection and Privacy Laws: GDPR, IT Act 2000 and Digital Personal Data Protection Act (DPDP), Intellectual Property Rights and Content Ownership in Social Networks, Cybercrime, Digital Evidence and Social Media Forensics, Ethical Challenges in Social Network Monitoring and Surveillance, Content Moderation and Platform Responsibility, Deepfakes and AI-Generated Content Threats, Emerging Trends: Artificial Intelligence, Blockchain and Decentralized Social Networks	**9**

Course Outcomes:

Course outcome	Descriptions
CO1	Understand the structure, types, and characteristics of social networks and apply graph theory concepts for social network analysis
CO2	Identify and analyze various security threats, attacks, and
	vulnerabilities specific to online social networking platforms
CO3	Design and implement privacy-preserving mechanisms, access control models, and anonymization techniques for social media security
CO4	Apply detection and prevention techniques for malicious activities including spam, phishing, malware propagation, and fake profiles in social networks
CO5	Evaluate ethical considerations, legal frameworks, and emerging technologies for securing social network ecosystems and analyze real-world case studies of social media security incidents

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1								2	1
CO2	3	3	2	2	1	1		1				3	3
CO3	3	3	3	2	2	2	1	2			1	3	3
CO4	3	3	3	3	2	1		1			1	3	3
CO5	2	3	2	2	1	3	2	3	2	2	2	2	2

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Security and Privacy in Social Networks	Yaniv Altshuler, Yuval Elovici, Armin B. Cremers, Nadav Aharony, Alex Pentland	Publisher: Springer Year: 2013 ISBN: 978-1-4614-4138-4 (Print), 978-1-4614-4139-1 (eBook)
2	Social Media Security: Leveraging Social Networking While Mitigating Risk	Michael Cross	Author: Michael Cross Publisher: Syngress (Elsevier) Year: 2013 ISBN: 978-1-597-49986-6

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Online Social Networks: Threats and Solutions	Michael Fire, Rami Puzis, Yuval Elovici	Publisher: IEEE Communications Surveys & Tutorials Year: 2014
2	Social Network Security: Issues, Challenges, Threats, and Solutions	Shailendra Rathore, Pradip Kumar Sharma, Vincenzo Loia, Young-Sik Jeong, Jong Hyuk Park	Publisher: Information Sciences (Elsevier) Year: 2017 Volume: 421 DOI: 10.1016/j.ins.2017.08.063

Web links and Video Lectures (e-Resources):
https://nptel.ac.in/courses/106106146
https://onlinecourses.nptel.ac.in/noc26_cs82/preview?utm_source=chatgpt.com
https://onlinecourses.nptel.ac.in/noc22_cs117/preview?utm_source=chatgpt.com
https://onlinecourses.nptel.ac.in/noc26_cs144/preview

Course Assessment and Evaluation: Regular Theory Courses

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Continuous Internal Evaluation (CIE): 50 Marks						
Sl.No	Components		Duration(min)	Marks Conducted for	Marks reduced to	40% Min Marks in each component to write SEE
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	

Total CIE	150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline), ……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice.
Each question may have a maximum of 4 sub-divisions. Minimum passing Marks: 40% of Max Marks

Department: CSE (Cyber Security)			Semester:	VII
Subject: Quantum Computing				
Subject Code:	22CY7PE63		L – T – P – S:	42-0-0-48 (In Hrs)
			Credits:	3

	Course Objectives
1	Explain the basic concepts of quantum computation and its physics
2	Use various operators of quantum computation and work on quantum transformation
3	Illustrate the working of some standard quantum algorithms
4	Analyse the complexities involved in working of quantum algorithms

Unit	Description	Hrs
I	**Quantum Building Blocks:** Quantum mechanics of Photon Polarization, Single Quantum bits, Single Qubit Measurement, A Quantum key Distribution Protocol, State Space of a Single-Qubit System, Direct Sums and Tensor Products of Vector Spaces, State Space of an n-Qubit System, Entangled States, Multi-Qubit Measurement, QKD using Entangled states.	09
II	**Multiple-Qubit States Measurements**: Dirac's bra/ket Notation for Linear transformations, Projection operators, Hermitian Operator Formalism, Bell's Theorem.	08
III	**Quantum State Transformations:** Unitary transformations, Simple Quantum Gates, Pauli transformations, Hadamard Transformations, Multiple-Qubit Transformations, Controlled-NOT and other singly controlled gates, Applications of Simple Gates, Dense coding, Quantum teleportation.	09
IV	**Quantum Algorithms**: Computing with Superpositions, Walsh-Hadamard transformation, Quantum Parallelism, Notions of Complexity, Query Complexity, Communication Complexity, Simple Quantum Algorithm.	08

V	**Simon's And Grover's Algorithm:** Classical reduction to Period-Finding, Example illustrating Simon's Algorithm, DLP and Hidden Subgroup Problems, Grover's Algorithm, Amplitude amplification.	**07**

Reference Books:

1	Quantum Computing: A Gentle Introduction, Eleanor Rieffel and Wolfgang Polak, 2011, The MIT Press, ISBN 9780262015066
2	An introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, Muchele Mosca, Oxford University Press, 2007, ISBN-13: 978 0198570493, ISBN-10: 019857049X
3	Quantum Computing for Computer Scientists, 1st Edition, Noson S. Yanofsky and Mirco A. Mannucci, Cambridge University Press, 2008, ISBN 978-0-521-879965
4	Quantum Computing for Everyone, Chris Bernhardt, MIT Press, 2019, ISBN: 9780262039253

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc26_cs89/preview?utm_source=chatgpt.com

https://onlinecourses.nptel.ac.in/noc26_cs06/preview?utm_source=chatgpt.com

https://onlinecourses.nptel.ac.in/noc26_ph15/preview

https://onlinecourses.nptel.ac.in/noc26_ph14/preview

Course Outcomes:

Course outcome	Descriptions
CO1	Explain the various essentials of quantum computation, Qubits, and Quantum operators
CO2	Analyse working of quantum transformations and quantum gates
CO3	Describe principle of working of some of the standard quantum algorithms and their applications
CO4	Investigate the applications of quantum computing and quantum cryptography

Course Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	-	1	-	1
CO2	3	2	2	1	-	-	-	-	-	1	-	1
C03	1	3	2	2	-	-	-	-	-	1	-	1
C04	3	2	3	2	-	-	-	-	-	1	-	1
CO5	2	3	3	2	-	-	-	-	-	1	-	1

Course Assessment and Evaluation: Regular Theory Courses

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Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	Components		Duration(min)	Marks Conducted for	Marks reduced to	40% Min Marks in each component to write SEE
1	Theory-Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL 1	-	25	5	4
		ABL 2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module
Activity Based Learning (ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program (1 week/online or offline), ……Etc

Semester End Examination (SEE): Max 100 Marks, Duration: 3 Hrs The question paper shall contain 10 full questions (20 marks each), with 2 questions from each of the 5 Modules, ensuring 100% syllabus coverage. Students shall answer any 5 full questions with 1:1 choice.
Each question may have a maximum of 4 sub-divisions. Minimum passing Marks: 40% of Max Marks

Department: CSE (Cyber Security)			**Semester:**	**VII**
Subject: PROJECT WORK PHASE – I				
Subject Code:	**22CYPW707**		**L – T – P - S:**	**0-0-28-32**
			Credit	**2**

Sl. No	Experiment Description
1	The students are divided into batches (same discipline or interdisciplinary), with each batch consisting of a minimum of 2 students and a maximum of 4 students (including bright and slow learners) for Major Projects,
2	Each batch must submit three different synopses in a particular domain of their interest within one week from the commencement of the semester.
3	Faculties are allocated to the project batches as guides, considering their area of specialization.
4	Students MUST get approval of the project title from the respective guide and the Department Project Evaluation Committee (DPEC).
5	After finalization of the project title and guide allotment, students should give a presentation to their respective guides.
6	Interdisciplinary projects must have a guide from each of the participating departments.
7	The students should maintain a project summary report consisting of the day-to-day work carried out by them, monitored weekly by the guide.
8	The Project work phase-I done by students, is evaluated for 50 CIE Marks No SEE

Course Outcomes:

Course outcome	Descriptions
CO1	**Apply** theoretical knowledge and engineering principles to design and implement a practical solution for a real-world problem.
CO2	Effectively **plan, manage, and execute** a multi-disciplinary project, adhering to timelines and resource constraints.
CO3	**Analyze, interpret, evaluate** experimental results and identify the areas for improvement.
CO4	**Communicate** complex technical information effectively through reports, presentations, and demonstrations.
CO5	**Demonstrate** ethical considerations, professional conduct, and an understanding of intellectual property rights through their team work.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	2	2	1		1	1	1	1		
CO2	2	2	3	1	2			1	2	2	2		
CO3	3	3	2	3	2			2	2	2			
CO4			2					3	2	3			
CO5								3	3	2	1		

Major Project Evaluation

Major Project work done by students is evaluated in two phases

a) **Phase 1** evaluation (CIE: 50 marks) No SEE
b) **Phase 2** evaluation (CIE: 50 marks + SEE: 50 marks)

a) Phase-1 CIE Evaluation (50 Marks) (During 7th Semester)

Evaluation Parameters	Marks
Synopsis evaluation by DPEC (Relevance of the topic, Novelty, societal need, Industry based)	10
Meeting regularity	10
Presentation Evaluation by DPEC	15
Viva-voce	5
Report writing	10
Total	**50**



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(A Constituent College of SSAHE, Accredited by NBA)
Scheme of Teaching and Examination for NEP 160 Credits (2022 Scheme)



8TH SEMESTER **ACADEMIC YEAR: 2026-27**

Sl. No.	Course Code		Course Title	Teaching Dept.	Teaching Hrs.				Total	Credits	Examination			
					L	T	P	S			CIE Marks	SEE Marks	Total Marks	Exam Hrs
1	PC	22CYTS801	Technical Seminar	CY	-	-	-	30	30	1	50	-	50	-
2	PC	22CYIN802	Internship	CY	-	-	30	30	60	2	50	50	100	3
3	PW	22CYPW803	Project Work Phase-2	CY	-	-	140	130	270	9	50	50	100	3
				Total	**0**	**0**	**170**	**190**	**360**	**12**	**150**	**100**	**250**	**-**

L: Lecture, T-Tutorial, P-Practical/Drawing, CIE: Continuous Internal Evaluation, SEE: Semester End Examination, S: Self Learning. XX: Branch Code

Credit Distribution: PW: Project Work-09, Technical Seminar-1, Internship-2, **Total=12 Credits**.

HS: Humanity & Social Sciences-11, BS: Basic Science-22,ES: Engineering Science-23,PC: Professional Core-66, PE: Professional Elective-18,OE: Open Elective-06, Project Work/Seminar/Internship =14,Total Credits=160.

CONVENER, CDC **DEAN (ACADEMICS)** **PRINCIPAL**

Department: CSE (Cyber Security)			Semester:	VIII
Subject: Technical Seminar				
Subject Code:	22CYTS801		L – T – P - S:	0– 0 – 0 – 30
			Credits	1

Sl. No	Guidelines for preparing Technical Seminar
1	**Selection of Topic/Domain:** Each student shall identify a topic/domain of current cutting-edge technology in the field of Computer Science and Engineering. Student is also allowed to choose the recent research paper/article of the standard journal.
2	**Approval of the Topic**: After selecting the topic or research paper, get approval by the concerned faculty.
3	**Collect and acquire thorough knowledge:** Students are required to collect sufficient information (past and recent updates) about the topic they have chosen and allowed to read the related research papers and refer the books.
4	**Final Report and Presentation**: Students are required to prepare a final report of the collected information about the topic chosen and prepare a Power point presentation under the guidance of the concerned faculty.

Course Outcomes:

Course outcome	Descriptions
CO1	**Identify** and **explore** recent advancements or emerging technologies relevant to the engineering domain.
CO2	**Analyze** and synthesize information from diverse technical literature sources.
CO3	**Develop** effective communication skills through preparation, delivery of technical presentations and dissertations.
CO4	**Demonstrate** the ability to work independently and exhibit self learning capabilities.
CO5	**Exhibit** professional ethics and understand the societal impact of modern technologies discussed in seminars.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2			2	2								
CO2	2	3		2	2									
CO3					2									
CO4									3		2			
CO5						2	2	3						

Technical Seminar:

Break-up of CIE marks for Technical Seminar

Evaluation Parameter	Marks
Relevance of the topic	10
Report	10
Presentation	20
Viva-Voce	10
Total	**50**

Department: CSE (Cyber Security)			**Semester:**	**VIII**
Subject: Internship				
Subject Code:	**22CYIN802**		**L – T – P - S:**	**0– 0 – 30 – 30**
			Credits	**2**

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Agalakote, B.H.Road, Tumkur – 572 107.KARNATAKA, INDIA.

No.SSAHE/ACA/49/2025 Date: 24/03/2025

NOTIFICATION

Internship Guidelines
(For UG Bachelor of Engineering Students – NEP 160 Credits Batch of Sri Siddhartha Institute of Technology)

Course Code : 22XX8IS4
Course Title : Internship (Audit Course)
Applicability: Compulsory for all NEP (160 credits) Batch Students

In accordance with the academic regulations, the following guidelines are established for the Internship (Audit Course), which is a mandatory requirement for all undergraduate B.E. students at Sri Siddhartha Institute of Technology *(A constituent college of Sri Siddhartha Academy of Higher Education, Deemed to be University u/s 3 of the UGC Act, 1956)*, Tumkur, Karnataka:

- **Course Overview:**
  - The Internship (Audit Course) is a mandatory requirement but carries "0" credits.
  - There is no Semester End Examination (SEE); evaluation is conducted solely through Continuous Internal Evaluation (CIE).
- **Internship Duration & Mode:**
  - Students must complete an internship (Hybrid Mode) for a minimum of 6 weeks and a maximum of 8 weeks (cumulative period).

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Course Outcomes:

Course outcome	Descriptions
CO1	**Apply** theoretical engineering knowledge to real-world industry problems and practices.
CO2	**Demonstrate** problem-solving and technical skills in a professional work environment
CO3	**Exhibit** professional ethics, teamwork, and interpersonal skills in a multidisciplinary setting.
CO4	**Communicate** effectively through technical reports, logs, and presentations based on internship experience
CO5	**Understand** industry practices, project management techniques, and business processes relevant to engineering

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	2	2						2		
CO2	2	3	3	3	3				3		2		
CO3			2			2	2	3	3	2			
CO4							1	2	3	3			
CO5			2		2	2	2		1	2	3		

Internship Evaluation

Break-up of CIE marks for Internship evaluation

Evaluation Parameter	Marks
Assessing the relevancy of the topic by the Dept. internship Evaluation Committee (Faculty supervisor and one senior professor)	10
Company/ Industry assessment	10
Report writing	10
Presentation	15
Viva- Voce	05
Total	**50**

Break-up of SEE marks for Internship evaluation

Evaluation Parameter	Marks
Report	**08**
Presentation	**35**
Viva-voce	**07**
Total	**50**

Department: CSE (Cyber Security)			**Semester:**	**VIII**
Subject: Project Work Phase-II				
Subject Code:	**22CYPW803**	**L – T – P - S:**	**0-0-140-130**	
		Credits:	**9**	

Sl. No	**Experiment Description**
1	Student should give a presentation to the guide on the system requirements and design phase which includes hardware and software requirements, system architecture, high level design, low level design, system models, required implementation tools and algorithms.
2	Students are allowed to complete the project by implementing/simulating to generate expected results under the guidance of project guide.
3	Students are required to prepare the final report of the project with the generated results, future scope and conclusions.
4	Students have to demonstrate the working model of the project to their respective evaluation panel.
	Evaluation: ➢ Continuous evaluation is done by respective evaluation panel based on the regularity, technical knowledge and competence, programming skills, communication skills, collaborative learning, demonstration skills and documentation skills of the students. **Demonstration:** ➢ Students have to demonstrate the working model of the project to their respective project guide and for External examiner in the final exam.

Course Outcomes:

Course outcome	Descriptions
CO1	**Apply** theoretical knowledge and engineering principles to design and implement a practical solution for a real-world problem.
CO2	Effectively **plan, manage, and execute** a multi-disciplinary project, adhering to timelines and resource constraints.
CO3	**Analyze, interpret, evaluate** experimental results and identify the areas for improvement.
CO4	**Communicate** complex technical information effectively through reports, presentations, and demonstrations.
CO5	**Demonstrate** ethical considerations, professional conduct, and an understanding of intellectual property rights through their team work.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	2	2	1		1	1	1	1		
CO2	2	2	3	1	2			1	2	2	2		
CO3	3	3	2	3	2			2	2	2			
CO4			2					3	2	3			
CO5								3	3	2	1		

Major Project Evaluation

Major Project work done by students is evaluated in two phases

a) **Phase 1** evaluation (CIE: 50 marks) No SEE
b) **Phase 2** evaluation (CIE: 50 marks + SEE: 50 marks)

b) Phase-2 CIE Evaluation (50 Marks) (During 8th Semester)

Evaluation Parameters	Marks
Midterm Presentation Evaluation by DPEC+ Viva- Voce	10
Weekly Meetings with Guide recording all the remarks in diary	10
Final Presentation & demonstration Evaluation by DPEC	10
Project related activities (Mandatory): (Publication in National/ International Journals, Technodea, KSCST project submission, Plagiarism report)	10
Report writing	10
Total	**50**

c) Phase-2 SEE Evaluation (50 Marks)

Evaluation parameters	Marks
Writing the abstract and technical details of Project	08
Project demonstration / Presentation of project	25
viva voce	07
Project Report	10
Total	**50**